

A2: Networks - Exam Lab



This exam practice page is designed for the **A2: Networks** unit of the new IB Computer Science syllabus (first assessment 2027).

Syllabus Links:

[A 2.1 Network fundamentals](#)

[A 2.2 Network architecture](#)

[A 2.3 Data Transmissions](#)

[A 2.4 Network security](#)

The Exam Scenario

Scenario: The "Smart-Care" Medical Complex. A large urban hospital has recently upgraded its digital infrastructure. The complex uses a high-speed **Local Area Network (LAN)** to connect surgical robots, patient monitoring systems, and administrative offices. To protect patient privacy while allowing doctors to access records remotely, the hospital employs a **Virtual Private Network (VPN)** and extensive **Network Segmentation**.

Core Questions (HL and SL)

Give these questions a try! If you're stuck, check out the 'Strategy' section to help you get started. Once you're finished, use the Mark Scheme and Worked Example to review your progress. If these feel a bit tough right now, head back to the eBook pages to sharpen your understanding first.

Question 1: Network Fundamentals & Protocols (SL & HL)

- (a) **Identify** one characteristic of a **Personal Area Network (PAN)** that distinguishes it from the hospital's **Local Area Network (LAN)**. **[2 marks]**
- (b) The hospital's website uses **HTTPS** for its patient portal. **Explain** the importance of the "Secure" aspect of this protocol compared to standard **HTTP**. **[3 marks]**

Level 1: Strategy

Level 2: Mark Scheme

Level 3: Worked Example

Question 2: Architecture & Segmentation (SL & HL)

- (a) The hospital uses **Virtual Local Area Networks (VLANs)** to separate the guest Wi-Fi from the surgical robot control system. **Explain** one reason why **network segmentation** is used in this context. **[3 marks]**
- (b) **Compare and contrast** the **client-server** and **peer-to-peer (P2P)** networking models for use in a hospital environment. **[4 marks]**

Level 1: Strategy

Level 2: Mark Scheme

Level 3: Worked Example

HL Only : Advanced Networking

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Question 3: Data Transmission & TCP/IP (HL Only)

- (a) **Describe** how the **Transport layer** and the **Internet layer** of the **TCP/IP model** interact to ensure a medical image is sent correctly across the network. **[4 marks]**

(b) The hospital's IT department must choose between **static routing** and **dynamic routing** for its emergency response tablets. **Outline** one advantage and one disadvantage of using **dynamic routing** in this scenario. **[4 marks]**

Level 1: Strategy

Level 2: Mark Scheme

Level 3: Worked Example

Question 4: Vulnerabilities & Servers (HL Only)

(a) A hospital server is targeted by a **Distributed Denial of Service (DDoS)** attack. **Explain** how this attack functions and **suggest** one countermeasure the hospital could implement. **[4 marks]**

(b) **Outline** the role of a **Proxy Server** in protecting the hospital's internal network from external threats. **[2 marks]**

Level 1: Strategy

Level 2: Mark Scheme

Level 3: Worked Example

How confident do you feel with this topic?

Exam Ready: I answered most questions correctly without looking at the eBook. I can explain these concepts to someone else and understand the "why" behind the mark scheme.

Move on to the next topic.

Practitioner: I understood the general concepts, but I missed a few marks on specific terminology or complex applications. I needed the "Strategy" guide for the harder questions.

Annotate your answers with the Mark Scheme and revisit the eBook sections where you lost marks. You can also use the bookmarking and notes feature on the eBook pages to highlight certain areas.

Novice: I struggled to get started on the questions. I found it difficult to apply the theory to the practice scenarios.

Go back to the eBook and Worked Examples before attempting these questions again. Perhaps use some of the Quizlets to practice the key terms and review the command terms.

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